

CAPSTONE PROJECT REPORT

Plant Disease Classification with Deep Learning and Transfer Learning

PlantVillage | SimpleCNN, ResNet-18 and ResNet-50 | Detailed Project Report

Team	Dushyant Kumar, Deepthi Prakash, Shahina Pathan, Venkata S. Sarma
Project Mentor	Zia Mahmood Hussain
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Executive Objective

Develop and evaluate a robust image-classification pipeline for 38 PlantVillage crop-disease classes. The project compares a custom SimpleCNN with ImageNet-pretrained ResNet-18 and ResNet-50, evaluates frozen-backbone and layer4+FC fine-tuning strategies, measures data efficiency, calibrates predictive confidence with temperature scaling, and uses Grad-CAM to inspect whether predictions are driven by leaf regions rather than background artifacts.

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1. Problem Statement

Plant disease recognition from leaf images is a multi-class visual classification problem in which a useful model must distinguish visually similar diseases, remain fair across imbalanced classes, generalise to held-out images, and provide interpretable and meaningful confidence estimates. A high aggregate accuracy alone is insufficient if minority classes are ignored, duplicate images leak across splits, or the network relies on background cues rather than disease symptoms.

Key Challenges

- 38-class classification with substantial variation in class frequency.
- Potential duplicate-image leakage between training, validation and test sets.
- Domain shift risk: PlantVillage images are controlled compared with field conditions.
- Need to compare learning from scratch with transfer learning rather than assume pretraining is superior.
- Need for trustworthy probabilities and visual explanation of model attention.

Proposed Problem Statement

Design, train and evaluate a deep-learning system for PlantVillage disease classification that combines duplicate-safe data splitting, field-oriented augmentation, class-balanced optimisation, transfer learning, data-efficiency analysis, confidence calibration and Grad-CAM interpretation. The principal performance target used in the notebook is test macro-F1 ≥ 0.88 .

2. Project Objectives and Scope

- Establish a SimpleCNN baseline trained from scratch.
- Benchmark ImageNet-pretrained ResNet-18 and ResNet-50.
- Compare frozen-backbone training with selective fine-tuning of layer4 + FC.
- Use macro-F1 alongside accuracy and loss because the class distribution is imbalanced.
- Measure performance at 15%, 30%, 45%, 60%, 75%, 90% and 100% of the training data.
- Evaluate the final ResNet-50 on the fixed held-out test set.
- Apply temperature scaling fitted only on validation logits and measure ECE before and after calibration.
- Inspect three correct and two incorrect predictions together with Grad-CAM.

3. Dataset Scope and Data Integrity

The notebook loads the PlantVillage color-image set with 54,305 images across 38 classes. Perceptual hashing (pHash) is used to identify visually duplicate images. Twenty-two duplicate rows are removed, leaving 54,283 images. Duplicate groups are used as the split unit so perceptually identical images cannot cross train, validation and test boundaries.

Stage	Images / Result
Raw dataset	54,305 images, 38 classes
Duplicate-group rows	44 rows in 22 duplicated hash groups
Cross-label duplicate groups	0
Duplicates removed	22
Clean dataset	54,283 images
Train split	37,998
Validation split	8,142
Test split	8,143

The split is approximately 70/15/15 and is stratified by label at the duplicate-group level. Explicit overlap checks returned zero train-validation, train-test and validation-test group overlap.

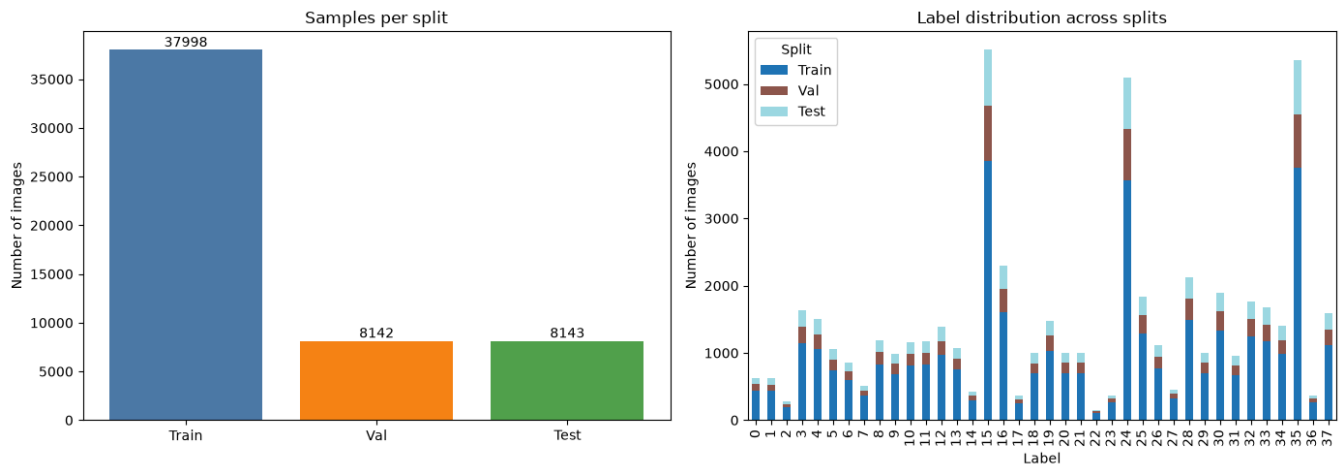


Figure 1. Dataset and class-distribution diagnostics from the project notebook.

Class Imbalance

The majority class is Orange____Haunglongbing_(Citrus_greening). A deliberately naive majority-class classifier achieves accuracy 0.101 but macro-F1 only 0.005, demonstrating why accuracy alone would hide poor minority-class performance.

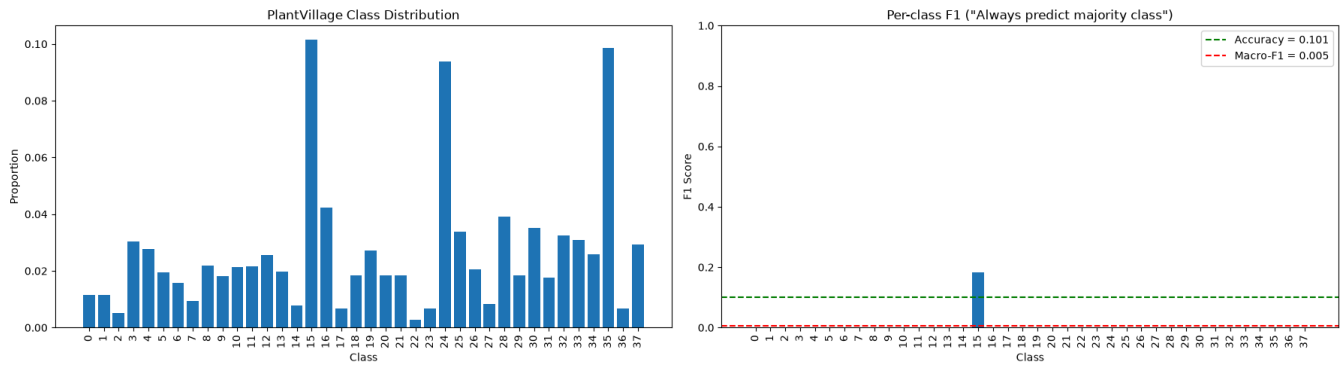


Figure 2. Class distribution and the failure of a majority-only classifier under macro-F1.

4. Technical Design Blueprint

The pipeline is modular: data indexing -> duplicate detection -> group-stratified splitting -> augmentation and normalisation -> weighted loss -> model training -> validation-based checkpoint selection -> held-out test evaluation -> data-efficiency analysis -> confidence calibration -> Grad-CAM interpretation.

Design Area	Implementation	Purpose
Input	RGB leaf images resized to 224x224	Standardised model input
Integrity	pHash duplicate grouping	Prevent split leakage
Training augmentation	Flip, +/-10° rotation, color jitter, blur, random erasing	Improve robustness to field-like variation
SimpleCNN normalisation	Training-set mean/std	Dataset-specific baseline preprocessing
ResNet normalisation	ImageNet mean/std	Match pretrained backbone expectations
Loss	Weighted cross-entropy	Compensate for class imbalance
Selection metric	Validation macro-F1	Balanced class-level model selection
Early stopping	Patience = 3	Limit unnecessary training
Interpretability	Grad-CAM on final layer4 block	Inspect spatial evidence
Calibration	Temperature scaling	Improve probability reliability

5. Model Architectures and Experiment Matrix

The custom SimpleCNN contains four convolutional blocks with Conv2d, BatchNorm2d, ReLU and max pooling, followed by adaptive average pooling and a linear classifier. The transfer-learning models replace the original ImageNet classifier with dropout (p=0.3) plus a 38-class linear layer.

Model	Pretrained	Trainable Region	Total Params	Trainable Params
SimpleCNN	No	All	0.25M	246,438
ResNet-18 Phase 1	Yes	FC only	11.20M	19,494
ResNet-18 Phase 2	Yes	Layer4 + FC	11.20M	8,413,222
ResNet-50 Phase 1	Yes	FC only	23.59M	77,862
ResNet-50 Phase 2	Yes	Layer4 + FC	23.59M	15,042,598

Transfer learning trains only the small replacement head in Phase 1

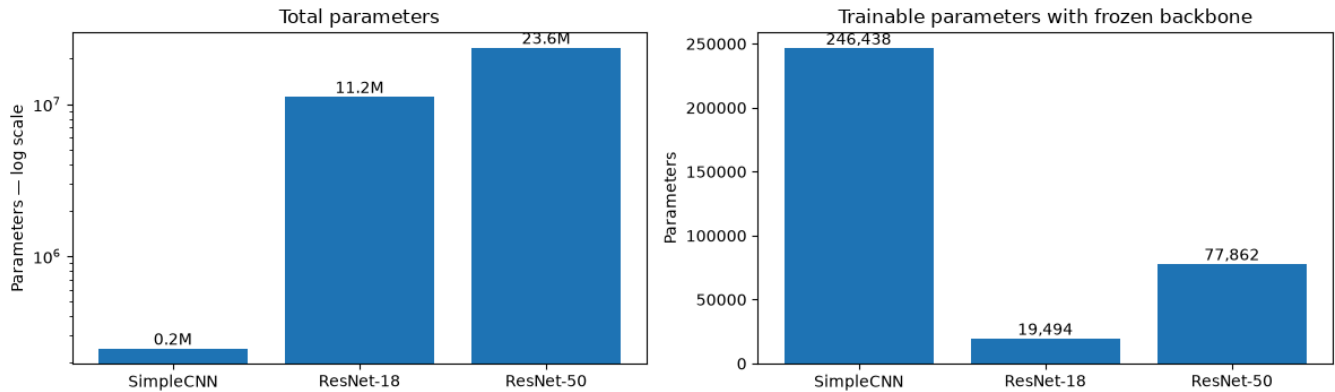


Figure 3. Parameter-count comparison across the candidate architectures.

6. Methodology

6.1 Preprocessing and Augmentation

For the SimpleCNN path, the training-set RGB mean is [0.4668, 0.4894, 0.4109] and standard deviation is [0.1992, 0.1749, 0.2174]. For pretrained ResNets, ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] are used. Training augmentation includes horizontal flipping, small rotations, color jitter, occasional Gaussian blur and random erasing; validation and test images receive only resizing, tensor conversion and normalisation.

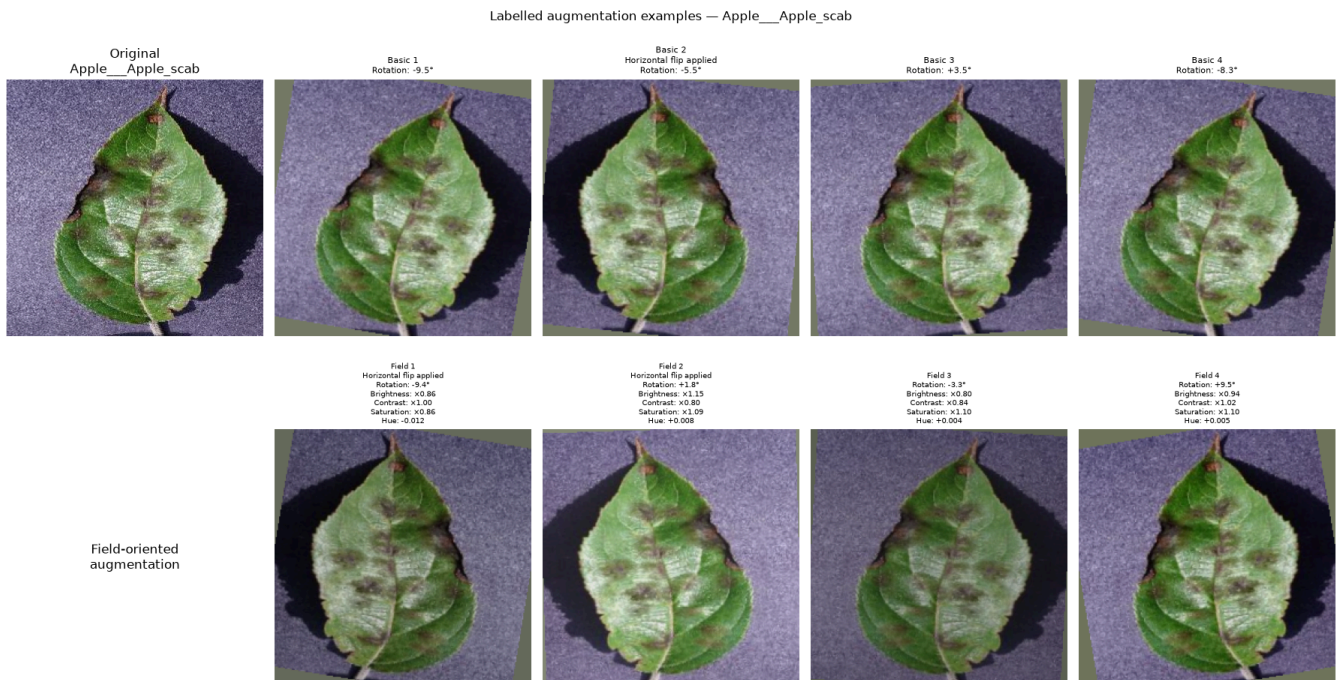


Figure 4. Examples of on-the-fly training augmentation.

6.2 Training Procedure

Models are trained with Adam and weighted cross-entropy. Each epoch records training and validation loss, accuracy and macro-F1. The best checkpoint is the epoch with the highest validation macro-F1. BatchNorm layers belonging to frozen parts are kept in evaluation mode during training so their running statistics are not updated.

Phase 1 trains only the replacement classification head at learning rate 1e-3. Phase 2 starts from the Phase-1 checkpoint, freezes the network globally, then explicitly unfreezes layer4 and FC; these trainable parameters are optimised at learning rate 1e-4. Thus Phase 2 is selective fine-tuning, not full-network fine-tuning.

7. Evaluation Metrics

Metric	Role
Cross-entropy loss	Optimisation objective and error magnitude
Accuracy	Overall fraction of correct predictions
Macro-F1	Equal-weight average F1 across all 38 classes; primary selection metric
Per-class precision / recall / F1	Class-specific error analysis
Confusion matrix	Pairwise disease-confusion analysis
ECE	Gap between confidence and empirical accuracy
Grad-CAM	Qualitative spatial localisation of prediction evidence

Macro-F1 is the principal comparison metric because every class contributes equally regardless of frequency. ECE uses 10 confidence bins and computes the sample-weighted absolute gap between mean confidence and observed accuracy.

8. Baseline and Transfer-Learning Results

Model	Val Loss	Val Accuracy	Val Macro-F1	Training Time (min)
SimpleCNN	0.2538	0.9193	0.9071	37.51
ResNet-18 (Frozen)	0.2017	0.9388	0.9249	20.33
ResNet-18 (Layer4 + FC)	0.0385	0.9891	0.9857	20.60
ResNet-50 (Frozen)	0.1565	0.9571	0.9455	22.81
ResNet-50 (Layer4 + FC)	0.0276	0.9921	0.9910	21.68

The transfer-learning models outperform the custom baseline while training faster. Fine-tuning layer4 + FC produces the largest improvement: ResNet-50 reaches validation macro-F1 0.9910 in this model-comparison run, compared with 0.9455 when only its head is trained.

SimpleCNN — prototyping subset
What to watch: loss goes DOWN, accuracy goes UP. A big train-val gap = overfitting.

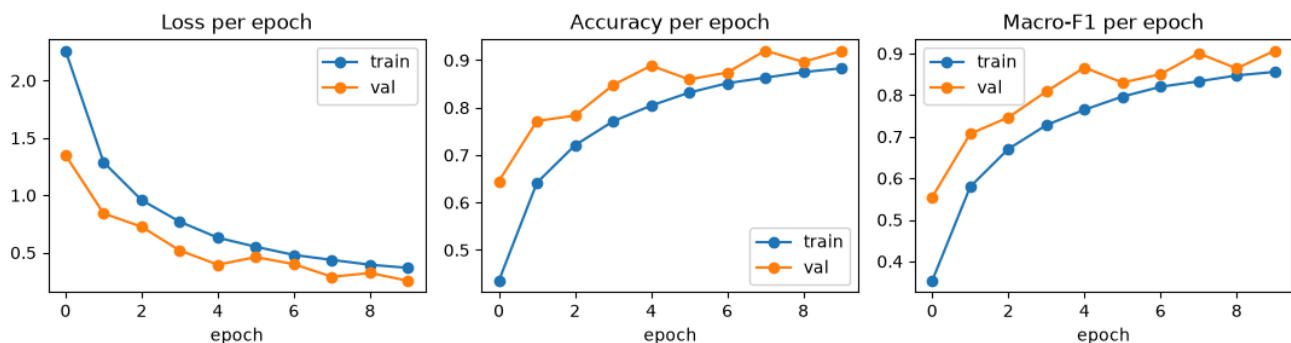


Figure 5. SimpleCNN learning curves.

ResNet-50 Phase 2 — unfrozen backbone
 What to watch: loss goes DOWN, accuracy goes UP. A big train-val gap = overfitting.

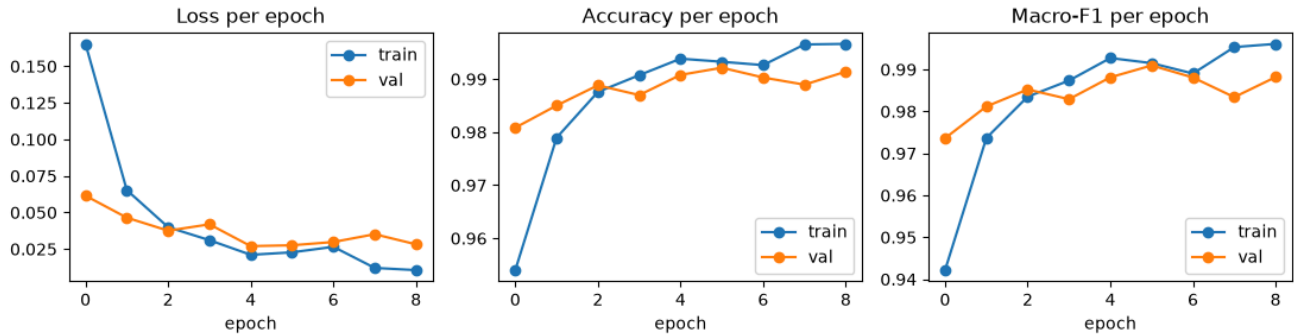


Figure 6. ResNet-50 Phase-2 learning curves.

9. Data-Efficiency Evaluation

To quantify how much labelled data the transfer-learning system needs, the training split is subsampled by duplicate group while the same validation and test sets remain fixed. Each fraction is trained in two stages: frozen-head training followed by layer4 + FC fine-tuning.

Train %	Train Images	Frozen Test Macro-F1	Fine-tuned Test Macro-F1
15%	5,699	0.914	0.968
30%	11,399	0.938	0.985
45%	17,099	0.945	0.989
60%	22,798	0.949	0.991
75%	28,498	0.955	0.992
90%	34,198	0.961	0.992
100%	37,998	0.959	0.996

Even at 15% of the training data, the fine-tuned ResNet-50 achieves test macro-F1 0.968, already above the 0.88 target. Performance rises quickly through 30-60% and then shows diminishing returns, reaching 0.996 at 100%.

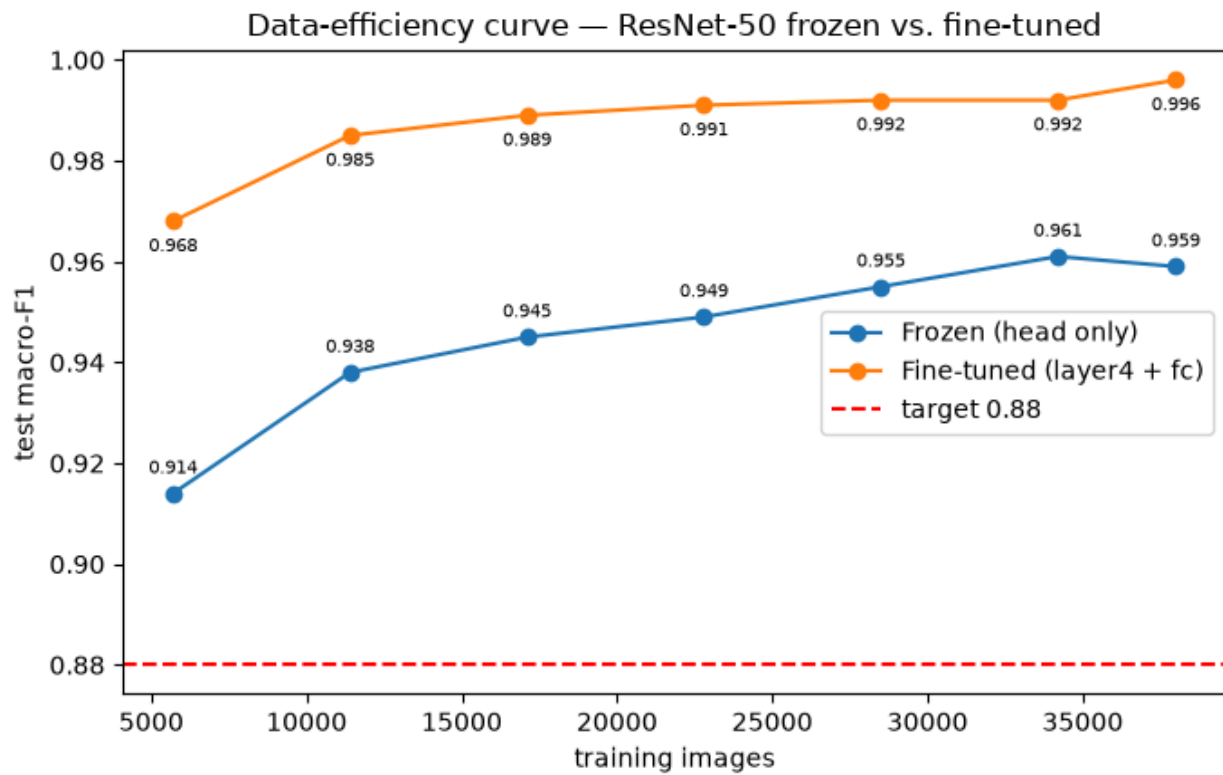


Figure 7. Data-efficiency curve for frozen and selectively fine-tuned ResNet-50.

10. Final Model Test Performance

The final model is the 100%-data ResNet-50 checkpoint after layer4 + FC fine-tuning. On the held-out test set of 8,143 images it achieves accuracy 0.997 and macro-F1 0.996, substantially exceeding the project target.

Metric	Final Result
Test samples	8,143
Test accuracy	0.997
Test macro-F1	0.996
Target macro-F1	≥ 0.88

Most classes achieve near-perfect F1. The classification report shows some remaining difficulty in closely related disease classes; for example, Corn Cercospora/Gray Leaf Spot has F1 0.957 and Corn Northern Leaf Blight has F1 0.976, while many classes reach 1.000.

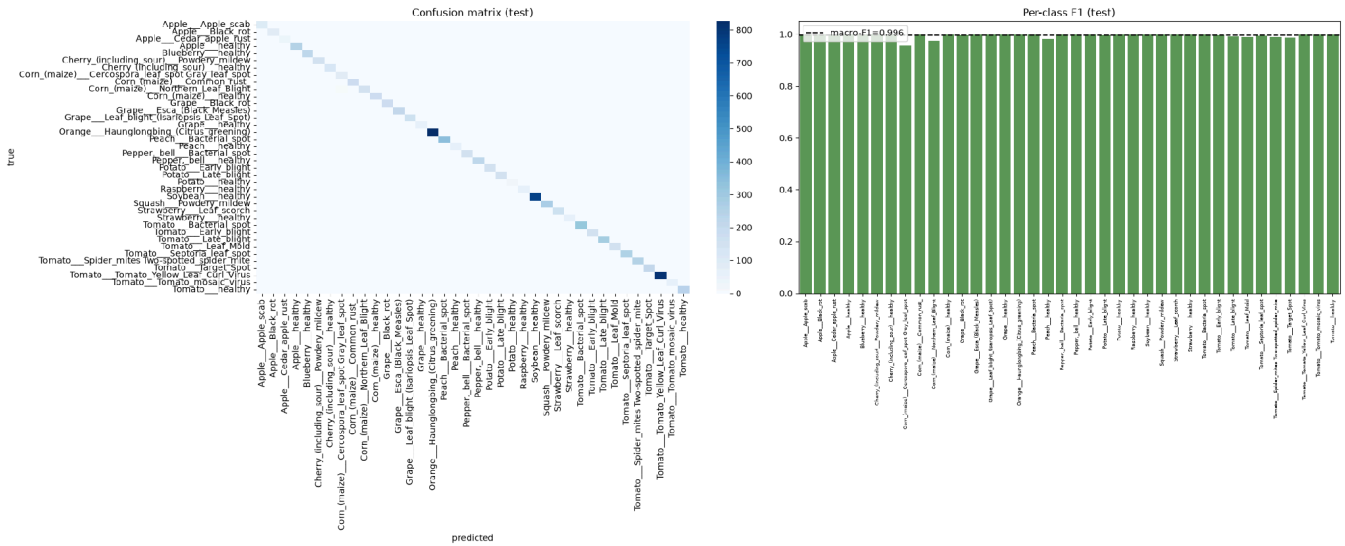


Figure 8. Final-model confusion matrix and per-class F1 analysis.

11. Confidence Calibration

Temperature scaling is fitted using validation logits only, preserving the test set for final assessment. The fitted temperature is $T = 0.615$. Because $T < 1$, the uncalibrated model is under-confident and scaling sharpens its probability distribution.

Calibration Metric	Before	After
Temperature T	-	0.615
Test ECE	0.049	0.004
ECE reduction	-	0.045

ECE decreases from 0.049 to 0.004 after temperature scaling. The reliability diagram therefore indicates that the calibrated probabilities align much more closely with observed correctness.

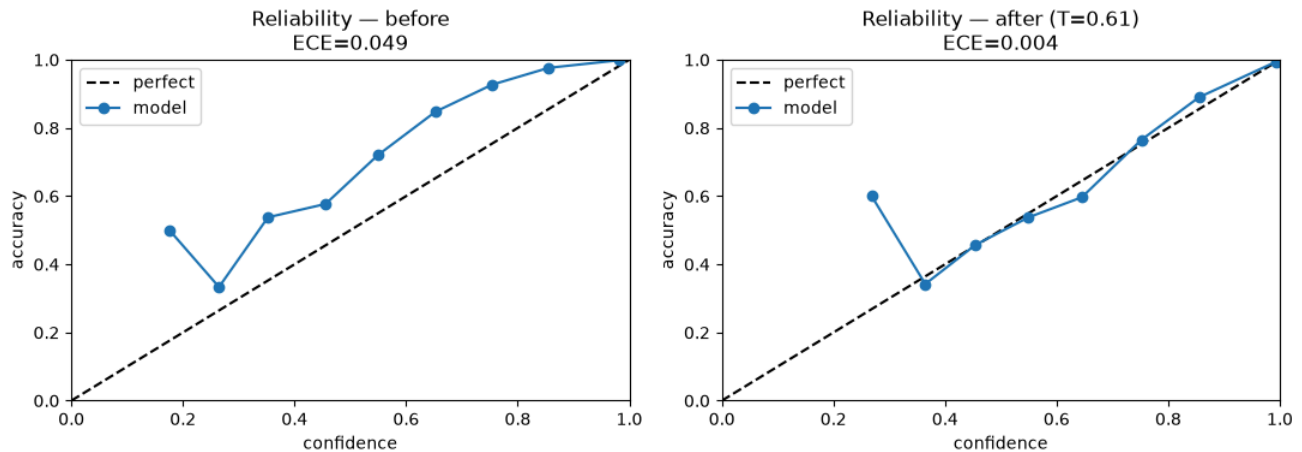


Figure 9. Reliability diagrams before and after temperature scaling.

12. Grad-CAM Disease Localisation

Grad-CAM is applied to the last block of ResNet-50 layer4. The report deliberately combines three correctly classified examples from different classes with two misclassified examples. This allows the explanation to be used not only as a success illustration but also as a failure-analysis tool.

The heatmaps should be interpreted qualitatively: activation concentrated on symptomatic leaf regions supports lesion-driven reasoning, while strong attention on image borders or background areas indicates possible shortcut learning. The two misclassified examples are especially useful for determining whether an error arose despite attention to the leaf or because the model focused on irrelevant context.

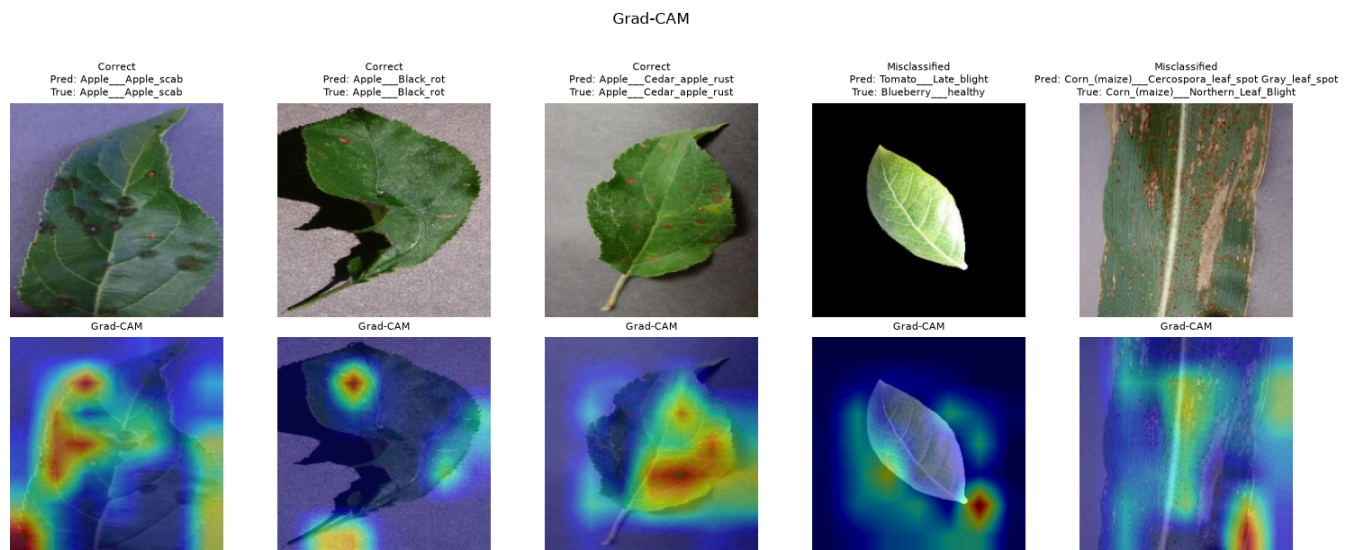


Figure 10. Grad-CAM comparison: three correct predictions from different classes and two misclassified cases.

13. Challenges and Mitigations

Challenge	Impact	Mitigation
Duplicate images	Inflated validation/test scores through leakage	pHash grouping, duplicate removal and group-level split
Class imbalance	Accuracy can favour frequent classes	Weighted cross-entropy and macro-F1
Controlled-image bias	Strong PlantVillage performance may not transfer directly to field photos	Field-oriented augmentation; future external field validation
Frozen BatchNorm	Running statistics could drift even when backbone weights are frozen	Keep frozen BatchNorm modules in eval mode
Overfitting during fine-tuning	High-capacity pretrained models can memorise	Validation macro-F1 checkpointing and early stopping
Unreliable confidence	Softmax score may not equal empirical correctness	Temperature scaling fitted on validation set
Background shortcut risk	Model may use setup/background cues	Grad-CAM inspection including misclassified cases

14. Key Findings

- Transfer learning clearly outperformed the SimpleCNN baseline: ResNet-50 Phase 2 reached validation macro-F1 0.9910 versus 0.9071 for SimpleCNN in the direct comparison.
- Selective fine-tuning of layer4 + FC was substantially better than training only the new classification head.
- ResNet-50 provided the strongest final performance, reaching test accuracy 0.997 and macro-F1 0.996 on 8,143 held-out images.
- Fine-tuned ResNet-50 was highly data-efficient: 15% of training data already produced test macro-F1 0.968.

- Temperature scaling materially improved confidence reliability, reducing ECE from 0.049 to 0.004.
- Grad-CAM extends evaluation beyond accuracy by showing where the network looks, including on two deliberately selected failure cases.
- Despite excellent in-dataset results, PlantVillage's controlled image conditions mean real-field generalisation remains the most important next validation step.

15. Conclusion and Future Work

This capstone produced an end-to-end plant-disease classification pipeline with explicit data-integrity controls, imbalance-aware training, baseline comparison, transfer learning, selective fine-tuning, data-efficiency experiments, probability calibration and visual interpretability. The final 100%-data ResNet-50 achieved 0.997 test accuracy and 0.996 macro-F1. The project therefore meets its stated performance target while also evaluating whether predictions are balanced across classes, probabilistically meaningful and spatially interpretable.

15.1 Completed Enhancements

- Duplicate-safe group splitting using perceptual hashes.
- Field-oriented augmentation and pretrained-model preprocessing.
- SimpleCNN, ResNet-18 and ResNet-50 comparison.
- Two-phase transfer learning with layer4 + FC fine-tuning.
- Seven-point data-efficiency curve.
- Final per-class test analysis.
- Temperature scaling and reliability diagrams.
- Grad-CAM with three correct and two misclassified examples.

15.2 Future Work

- Evaluate on independent field datasets and farmer-captured smartphone images.
- Test robustness to severe lighting changes, clutter, occlusion, multiple leaves and mixed symptoms.
- Report uncertainty-aware abstention thresholds using the calibrated probabilities.
- Perform targeted error analysis for the lowest-F1 disease pairs.
- Compare additional efficient backbones for mobile deployment.
- Quantify Grad-CAM localisation against lesion annotations when such masks are available.
- Measure inference latency, memory footprint and model size for deployment.

16. References

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End of Report